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Inventor(s)	Wynalda, Jr.; Robert M.

Mounting brackets and utility mount

Abstract

A selectively adjustable and lockable utility mount is used to mount a device to a stable structure. The utility mount includes a hydraulic lock that applies the locking force that maintains the locked condition of the utility mount. Mounting brackets for utility mounts have spaced-apart mount elements that engage a surface of a stable structure. Some configurations of mounting bracket include mount elements that are movable between stored conditions and deployed conditions. The stored conditions reduce the size of the device to make it more convenient to pack. A climbing stick uses the storable mount elements.

Inventors:	Wynalda, Jr.; Robert M. (Comstock Park, MI)
Applicant:	Fourth Arrow, LLC (Comstock Park, MI)
Family ID:	1000008779219
Assignee:	Fourth Arrow, LLC (Comstock Park, MI)
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Primary Examiner: Wujciak; Alfred J

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application claiming priority to U.S. patent application Ser. No. 18/807,526 filed Aug. 16, 2024, which application is a continuation application claiming priority to U.S. patent application Ser. No. 17/168,878 filed Feb. 5, 2021, U.S. Pat. No. 12,063,926; the disclosures of both are incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

1. Technical Field

(1) The present disclosure generally relates to mounting brackets and utility mounts for mounting or supporting items. The mounting brackets include spaced-apart, pointed feet that can grip, engage, or dig into a stable support surface to help secure the mount at a location. Some configurations of the feet are movable between stored and deployed positions. Exemplary mounts are disclosed in combination with a climbing stick and as receiving a utility mount. The disclosure also provides a utility mount configuration that uses a hydraulic lock to supply the locking force. The utility mount can be used to support a camera, a camera arm, a shooting rest, a shooting rest

arm, or other devices that need to be mounted to a stable structure.

2. Background Information

(2) Utility mounts for mounting equipment to a stable structure in an outside hunting location are known in the art. The devices and methods disclosed below provide improvements and new configurations to mounting brackets and utility mounts.

SUMMARY OF THE DISCLOSURE

(3) The disclosure provides a selectively adjustable and lockable utility mount that is used to mount a device to a stable structure. The utility mount includes a hydraulic lock that applies the locking force that maintains the locked condition of the utility mount. Exemplary devices to be mounted include a camera, a camera arm and camera combination, a shooting rest, or a shooting rest arm and shooting rest combination. Other devices include range finders, night vision gear, binoculars, gear holders, bow rests, or other equipment used when hunting, taking photos, or exploring outside. Exemplary stable structures include tree trunks, tree branches, fence posts, or telephone poles. The utility mount also can be used inside and mounted to stable structures in the nature of poles, beams, walls, furniture and the like.

(4) The disclosure provides different configurations of mounting brackets with spaced-apart mount elements that engage a surface of a stable structure to provide a stable configuration for the mounting brackets. Some configurations of the brackets are secured with straps that wrap around the stable structure. Some configurations of mounting bracket include mount elements that are movable between stored conditions and deployed conditions. The stored conditions reduce the size of the device to make it more convenient to pack. Embodiments of the mount elements have pointed engagement ends to grip the mount surface. The stored conditions cover the pointed engagement ends of the mount elements to prevent or limit damage to other items from the pointed engagement ends.

(5) The disclosure provides a mounting bracket with storable mount elements.

(6) The disclosure provides configurations wherein the mount elements are curved, pointed, and claw-shaped.

(7) The disclosure provides a mounting bracket that can be slid onto a strap loop without requiring the strap to be fed through an opening defined by the mounting bracket.

(8) The disclosure provides a mounting bracket in the form of a climbing stick that can be used to climb up onto a structure. The climbing stick includes upper and lower storable mount elements that pivot out to deployed conditions to bite into the structure to which the stick is mounted.

(9) The preceding non-limiting features of the disclosures, as well as others, are more particularly described below. A more complete understanding of the devices and methods can be obtained by reference to the accompanying drawings, which are not intended to indicate relative size and dimensions of the assemblies or components thereof. In those drawings and the description below, like numeric designations refer to components of like function. Specific terms used in that description are intended to refer only to the particular structure of the embodiments selected for illustration in the drawings, and are not intended to define or limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a view of a climbing stick with storable mount elements in the stored condition.

(2) FIG. 2 is a view of the climbing stick of FIG. 1 mounted to a tree with the mount elements in the deployed condition.

(3) FIG. 3 is a view of a first configuration of a utility mount mounting bracket with fixed mount elements.

(4) FIG. 4 is a view of an adjustable utility mount carried by the mounting bracket of FIG. 3.

- (5) FIG. 5 is a view of a second configuration of a utility mount mounting bracket with storable mount elements in the stored condition.
- (6) FIG. 6 is a view of the FIG. 5 configuration with the mount elements in the deployed condition.
- (7) FIG. 7 is a view of a third configuration of a utility mount mounting bracket.
- (8) FIG. 8 is a view of a third configuration of a utility mount mounting bracket with storable mount elements in the stored condition.
- (9) FIG. 9 is a view of the FIG. 8 configuration with the mount elements in the deployed condition.
- (10) FIG. 10 is an exploded view of the utility mount.
- (11) FIG. 11 is a view depicting a camera and camera arm carried by the utility mount.
- (12) FIG. 12 is a view depicting a shooting rest and arm carried by the utility mount.

DETAILED DESCRIPTION OF THE DISCLOSURE

- (13) An exemplary configuration of a climbing stick is indicated generally by the reference numeral 2 in FIGS. 1 and 2. Climbing stick 2 is a portable climbing device similar to a ladder that provides access to elevated locations. Climbing stick 2 generally includes a climbing stick body 4 that is elongated and has a first end portion 6, a second end portion 8, a front side, a rear side, and left and right sidewalls. The length of first 6 and second 8 end portions can be up to one third of the length of climbing stick body 4. Climbing stick body 4 can be a solid member, a rectangular or rounded hollow tube, a C or U channel, or an H beam or an I beam. Body 4 can be made from a metal, a polymer, a wood, or other rigid material. Climbing stick body 4 defines a strap opening 10 that receives a mounting strap 12 that passes through body 4 and around a stable structure 14 to secure climbing stick 2 in a useful position. Strap 12 can include a clasp or a ratchet that allows the user to tighten strap 12 around structure 14. Strap opening 10 is disposed within first end portion 6 or closer to first end portion 6 than second end portion 8. Stable structure 14 is typically a tree trunk but also can be a tree branch, a pole, or another structure the user wishes to climb.
- (14) Climbing stick 2 also includes a plurality of steps 20 that alternate on the different sides of climbing stick body 4. Each step 20 can move between a stored condition as shown in FIG. 1 to a deployed condition as shown in FIG. 2. Each step 20 includes a stop that engages the side of stick body 4 to position the step substantially perpendicular to stick body 4. The steps are crenulated for gripping the user's foot. Folding steps 20 to the stored condition makes climbing stick 2 smaller in profile to pack for carrying.
- (15) Climbing stick 2 further includes first 30 and second 32 mount elements. First mount element 30 is connected to first end 6 of climbing stick body 4. First mount element 30 is movable between a stored condition as shown in FIG. 1 and a deployed condition as shown in FIG. 2. Second mount element 32 is connected to second end 8 of climbing stick body 4. Second mount element 32 is movable between a stored condition as shown in FIG. 1 and a deployed condition as shown in FIG. 2. Mount elements 30 and 32 are connected with a fastener such as a nut and bolt combination or with a pivot pin. Each mount element 30 and 32 has at least one pointed engagement end 34 that can bite into a surface such as a tree trunk or a telephone pole to secure climbing stick 2 in place through cooperation with the strap 12. Each point comes to a tapering or sharpened point suitable for engaging stable surfaces for which stick 2 is configured to be used. For example, engagement ends 34 do not have to be as sharp for soft barks and woods compared to use on an aluminum pole. When in the deployed condition, engagement ends 34 point rearward with respect to climbing stick body 4 as shown in FIG. 2.
- (16) In the exemplary configuration, each mount element 30 and 32 includes two spaced-apart side members 40 connected together with a mount connector 42. Side members 40 can be curved, claw-shaped members. Each side member 40 has a pointed engagement end 34 such that each mount element 30 and 32 has a pair of spaced pointed engagement ends 34. When mount elements 30 and 32 are disposed in the stored condition, portions of climbing stick body 4 are disposed between side members 40 and between engagement ends 34 to protect the user and other equipment. When in the deployed condition, mount connectors 42 are disposed against portions of climbing stick body 4 to

function as stops to place mount elements **30** and **32** in the correct position for use.

(17) A first configuration of a mounting bracket for a utility mount is indicated generally by the reference numeral **100** in FIGS. **3** and **4**. Mounting bracket **100** is configured to mount a utility mount **102** in a stable configuration to a stable structure **14**. Mounting bracket **100** is secured to stable structure **14** with strap **12** which may be tightened with a ratchet. Mounting bracket **100** includes a plurality of mount elements **104** that have engagement ends **106** that engage stable structure **14** and can bite into structures such as wood to increase the stability of mounting bracket **100**. Each mount element **104** extends rearwardly from mounting bracket **100**. Each engagement end **106** comes to a tapered point suitable for engaging stable surfaces for which mounting bracket **100** is configured to be used. In the exemplary configuration, the edges of mount elements **104** define a curved taper like a claw. The point can be in two or three dimensions with the two dimension version (constant thickness) depicted in the drawings.

(18) Mounting bracket **100** generally includes a front wall **110**, sidewalls **112**, and a mount **114** that defines an opening **116** adapted to receive a portion of utility mount **102**. Front wall **110** is solid in the exemplary configuration. Other configurations of front wall **110** include fastener openings or openings to reduce the weight of front wall **110**. Further configurations of front wall **110** are made up of a plurality of elements that cooperate together to define front wall **110** such as a plurality of spaced bars or beams or a plurality of overlapping elements. These provide a structure to which utility mount **102** is fastened and from which sidewalls **112** project rearwardly. A projection **118** extends upwardly from a portion of front wall **110**. Two mount elements **104** are disposed above (and rearwardly of) the top of front wall **110** and two mount elements **104** are disposed below (and rearwardly of) the bottom of front wall **110**. Each sidewall **112** has one upper and one lower mount element **104** disposed in the same reference plane as the sidewall **112**. In a manner similar to front wall **110**, each sidewall **112** can be made up of different elements that project rearwardly from front wall **110** in order to support mount elements **104**. Sidewalls **112** can define openings and do not have to be continuous, flat plates.

(19) Mounting bracket **100** defines strap openings **120** between front wall **110** and sidewalls **112**. Strap openings **120** are closer to the top of front wall **110** than its bottom.

(20) Utility mount **102** is connected to mounting bracket **100** in FIG. **4** with a portion of utility mount **102** extending through opening **116**. In this configuration, a lock plate **130** extends between a portion of the top of utility mount **102** and the top of mounting bracket **100**. In this configuration, the bottom rear portion of lock plate **130** defines a slot **132** (FIG. **8**) that receives the upper end of front wall **110**. Within slot **132**, lock plate **130** defines a recess that receives projection **118** to limit lateral movement of utility mount **102**. In other configurations, utility mount **102** can be connected to mounting bracket **100** with fasteners (see FIGS. **5-6** for examples).

(21) FIGS. **5-6** depict a second exemplary configuration of a mounting bracket which is indicated generally by the reference numeral **200**. Mounting bracket **200** shares elements with mounting bracket **100** and the same reference numerals are used to identify these features. In this configuration, front wall **110** defines holes so that utility mount **102** can be secured to mounting bracket **200** with fasteners **202** such as machine screws. When fasteners are used, mounting bracket **200** is not required to include mount **114**. Connected the two together is desirable when the user takes both utility mount **102** and mounting bracket **200** with him when he leaves a hunting location. To make such transport more convenient, mounting bracket **200** includes mount elements **204** that are selectively movable with respect to sidewalls **112** between stored (FIG. **5**) and deployed (FIG. **6**) conditions. When in the stored configuration, engagement ends **206** of mount elements **204** are disposed between sidewalls **112** for protection against snagging in the user's pockets or pack. In addition, the overall size of the combination is reduced so that it occupies less space during transport.

(22) Mounting bracket **200** includes four storable mount elements **204** including two upper mount elements and two lower mount elements. Each mount element **204** is connected to a portion of one

of sidewalls **112** with a fastener **208** that allows mount element **204** to move between its stored and deployed conditions. When in the deployed condition, all engagement ends **206** point rearwardly with respect to sidewalls **112** as shown in FIG. **6**. Each mount element **204** includes a stop **210** that engages an end of front wall **110** to stop rotation of mount element **204** when mount element **204** is in the deployed condition. When in the stored conditions, two of mount elements **204** are rotated in toward front wall **110** until engagement ends **206** touch the rear surface of front wall **110**. The other two mount elements **204** are rotated on top of those. When stored, all four engagement ends **206** are disposed between sidewalls **112** to keep them out of the way.

(23) In the deployed condition of FIG. **6**, the two upper mount elements **204** are disposed above (and rearwardly of) the top of front wall **110** and the two lower mount elements **204** are disposed below (and rearwardly of) the bottom of front wall **110**.

(24) A third exemplary configuration of a mounting bracket is indicated generally by the reference numeral **300** in FIG. **7**. Mounting bracket **300** shares elements with mounting brackets **100** and **200** and the same reference numerals are used to identify these features. In this configuration, sidewalls **112** define slots **302** that allow the user to slide mounting bracket **300** over or off of a mounting strap **12** that is already installed. This allows the user to install and remove mounting bracket **300** without threading strap **12** through openings **120**. Each slot **302** has an upper closed end and an open lower end.

(25) A fourth exemplary configuration of a mounting bracket is indicated generally by the reference numeral **400** in FIGS. **8-9**. Mounting bracket **400** shares elements with mounting brackets **100** and **200** and the same reference numerals are used to identify these features. Mounting bracket **400** includes an upper handle **402** that extends between upper mount elements **204** and a lower handle **404** that extends between lower mount elements **204**. Handles **402** and **404** allow the user to move two mount elements at the same time and provides a grip for the user when carrying mounting bracket **400**. Each handle **402** and **404** is U-shaped and is connected at stops **210**.

(26) An exploded view of exemplary hydraulic utility mount **102** is shown in FIG. **10**. Utility mount **102** is used to hold or mount a device to a stable structure **14** such as a portion of a tree and maintain the device in a desired position during use. The position of the device being held can be adjusted by unlocking utility mount **102**, adjusting the position of device being held by utility mount **102**, and then re-locking utility mount **102**. This configuration of utility mount **102** does not require a tool to be locked and unlocked. The user can lock and unlock utility mount **102** by turning a finger knob **502**. Finger knob **502** is connected to a plunger **504** which is threaded to a mount body **506** (or a threaded insert **508**) which allows plunger **504** to move in and out with respect to mount body **506**. The inner end of plunger **504** engages hydraulic fluid disposed in a chamber between the inner end of plunger **504** and the inner end of a piston **510**. Moving plunger **504** inwardly by clockwise rotation to an engaged condition pushes the hydraulic fluid against piston **510** causing the outer end of piston **510** to push against device mount **512** to lock device **512** in place against an inner surface of a cup ring **514**. Turning finger knob **502** counterclockwise places plunger **504** in a disengaged condition and releases the pressure on the hydraulic fluid so that the position of device mount **512** can be adjusted.

(27) Adjustable utility mount **102** is used to support items such as an articulating arm **520** (FIGS. **11** and **12**), a camera **522**, a camera holder (or combination of both), a stick, a shooting rest **524**, a motion detector, or a light. Adjustable utility mount **102** can also be used to support camouflage, used directly as a gun rest, used as an equipment holder, or to support other gear.

(28) Finger knob **502** can be connected to or integral with plunger **504**. Finger knob **502** can be fixed to plunger **504** or removable and replaceable. In the exemplary configuration, finger knob **502** allows the user to adjust the position of plunger **504** without using a tool. In other configurations, finger knob **502** is not used and the outer end of plunger **504** is configured to receive a tool such as a screwdriver or a wrench which are used to adjust the position of plunger **504** with respect to mount body **506**.

(29) Plunger **504** includes an elongated threaded portion that threadedly engages mount body **506** directly or threadedly engages a threaded insert **508** that is fastened to mount body **506**. Insert **508** can be removed and replaced from mount body **506**. Insert **508** allows plunger **504** to be threaded into the insert **508** from the body-facing side of insert **508**. The inner end portion of plunger **504** which defines the plunger seal seats **526** and carries seals **528** is then slid into a plunger cavity that is defined by mount body **506**. Seals **528** carried by plunger **504** engage the inner surface of mount body **506** that defines the plunger cavity to prevent hydraulic fluid from exiting the fluid chamber of mount body **506** past plunger **504**.

(30) Mount body **506** can be secured to mounting bracket **100** by inserting its lower portion into opening **116** and using lock plate **130** as described above. A lower clamp **530** is used to tighten mount body **506** to mounting bracket **100**. Alternatively, mount body **506** can be secured with fasteners **202**.

(31) A cup ring **514** is removably connected to mount body **506** to allow a ball end **532** of device mount **512** to be positioned between the inner surface of cup ring **514** and the forward or outer surface of the front of piston **510**. Cup ring **514** can be connected with fasteners or with an interlocking mount configuration such as a bayonet-style connection. When installed as shown in FIG. 4, the forward or outer surface of ball end **532** engages the inner surface of cup ring **514** and the forward or outer surface of piston **510** engages the rear or inner concave surface of ball end **532**. The inner surface of cup ring **514** can be curved to correspond to the outer curved surface of ball end **532**. Similarly, the concave inner surface of ball end **532** can be curved to correspond to the curved outer surface of piston **510**. When plunger **504** is moved inwardly with respect to mount body **506**, the fluid pushes piston **510** against ball end **532** which is then trapped against cup ring **514** to lock the position of device mount **512**. Cup ring **514** defines a large opening that allows device mount **512** to swivel with respect to mount body **506**. Device mount **512** can rotate 360 degrees. The swivel movement can be combined with the rotation to allow the position of device mount **512** to be fully adjustable.

(32) Device mount **512** is provided in different configurations for different devices to be mounted. In the exemplary configuration, device mount **512** includes a stem that projects forward from ball head **532** and a cylinder that is connected at a right angle to the stem. An insert **540** is slid into the cylinder and can be locked in place with a thumb screw **541**. Insert **540** is configured to receive cylindrical mounts **542** (see FIGS. 11 and 12) from various items. In other configurations, device mount **512** can carry a standard threaded camera connector (1/4-20 UNC or 3/8-16 UNC thread) so that a camera can be connected directly to device mount **512**. In a further configuration, device mount **512** carries an adjustable clamp that engages the sides of a smart phone or tablet computer. Device mount **512** can carry a level **544**.

(33) Mount body **506** includes inner **550** and outer **552** concentric rings projecting forward from the front face of mount body **506**. A ring-shaped channel is defined between rings **550** and **552** to receive the rear portions of ball head **532** when device mount **512** is tilted with respect to mount body **506**. Cup ring **514** is mounted to outer ring **552** and can have the same outer and inner diameters so that the two match.

(34) The inner surface of inner ring **550** can define at least a portion of the cylinder in which the rear end of piston **510** slides and seals. The front of mount body **506** also can define a portion or all of the cylinder. The rear end of piston **510** defines a seal groove and carries a seal that engages the surface that defines the cylinder to seal the fluid in the mount body **506**. The hydraulic fluid is carried by mount body **506** in fluid communication between piston **510** and plunger **504**. Mount body **506** includes a sealable fluid port opposite the plunger opening where the fluid can be added or removed as needed.

(35) In the foregoing description, certain terms have been used for brevity, clearness, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly

construed. Moreover, the description and illustration of the invention is an example and the invention is not limited to the exact details shown or described. Modifications and alterations of those embodiments will be apparent to one who reads and understands this general description. The present disclosure should be construed as including all such modifications and alterations insofar as they come within the scope of the appended claims or equivalents thereof. Throughout the description and claims of this specification the words “comprise” and “include” as well as variations of those words, such as “comprises,” “includes,” “comprising,” and “including” are not intended to exclude additives, components, integers, or steps.

Claims

1. A utility mount assembly comprising: a mount body comprising a fluid chamber containing a fluid therein; the mount body defining a first opening extending into the chamber; the mount body defining a second opening extending into the chamber; the mount body defining a piston seat at the second opening; a piston dimensioned to be received by the seat and defining a spherical-shaped portion; a device mount having a first end; the first end having an outer surface and a concave inner surface corresponding to the spherical-shaped portion of the piston; a stop that is engaged by the outer surface of the first end of the device mount; and a resistance control that includes a plunger extending into the first opening and operable to compress the fluid in the chamber to variably modulate friction between the spherical-shaped portion of the piston and the concave inner surface of the first end of the device mount.
 2. The assembly of claim 1, wherein the stop is ring-shaped.
 3. The assembly of claim 2, wherein the outer surface of the first end of the device mount is curved and the stop has a curved surface; the curved outer surface of the first end of the device mount engaging the curved surface of the stop.
 4. The assembly of claim 1, wherein the mount body defines a recess aligned with the first end of the device mount; the position of the device mount being selectively adjustable with respect to the mount body to locate portions of the first end of the device mount within the recess.
 5. The assembly of claim 4, wherein the recess is ring-shaped.
 6. The assembly of claim 5, wherein the first end of the device mount has a ring-shaped edge.
 7. The assembly of claim 1, wherein the plunger includes a threaded portion and wherein the resistance control includes a body defining a threaded opening configured to rotatably engage the threaded portion of the plunger, and the mount body defines a depression for receiving the body, wherein a knob is attached to an end of the plunger and is operable to extend or retract the plunger relative to the body of the resistance control and into or out of the chamber, respectively.
 8. The assembly of claim 1, wherein the plunger is threaded to an insert that is removable from the mount body; the plunger being rotatable between engaged and disengaged positions.
 9. The assembly of claim 1, wherein the stop is a continuous ring-shaped body that is removably connected to the mount body to trap the first end of the device mount between the piston and the stop.
 10. The assembly of claim 9, wherein the first end of the device mount has a rounded outer surface that engages a concave curved inner surface of the stop.
 11. The assembly of claim 1, further comprising a lower clamp connected to the mount body; the lower clamp adapted to connect the mount body to a mounting bracket.
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